43rd AIAA Aerospace Sciences Meeting and Exhibit
10 - 13 January 2005, Reno, Nevada

AIAA 2005-1425

Experimental Investigation of Temperature and Species Concentration Characteristics of Swirling Spray Combustion

A. M. EL-Kady and S-M. Jeng

University of Cincinnati
Cincinnati, OH

H. C. Mongia

GE Transportation
Cincinnati, OH

43rd Aerospace Sciences Meeting & Exhibit
10-13 January 2005
Reno, Nevada

For permission to copy or republish, contact the American Institute of Aeronautics and Astronautics
1801 Alexander Bell Drive, Suite 500, Reston, VA 20191

Copyright © 2005 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved.

Experimental Investigation of Temperature and Species Concentration Characteristics of Swirling Spray Combustion

Ahmed M. EL-Kady* and San-Mou Jeng†
University of Cincinnati, Cincinnati, OH, 45220
Hukam Mongia‡
GE Transportation, Cincinnati, OH

The present work studies the effect of utilizing counter swirling flow on pollutant emission characteristics at different equivalence ratios and inlet air temperatures. A multiple gas analyzer based on Fourier transform infrared (FTIR) spectroscopy is employed to measure emissions concentrations generated during combustion of Jet-A fuel in a swirl cup assembly. These measurements are located within the combustion chamber and at the exit plane. The goal of this study is to provide more insight into nitric oxide and carbon monoxide formation/destruction processes within the recirculation zone and throughout the combustor. The results illustrate that NO_x production is a function of both the recirculation zone and the flame length. At low overall equivalence ratios, the recirculation zone is found to be the main producer of NO_x. At near stoichiometric conditions, the post recirculation zone appears to be responsible for the majority of NO_x produced.

Nomenclature

ϕ	=	Equivalence ratio
D	=	Swirler Flare exit diameter
FTIR	=	Fourier Transform Infra-Red
LPM	=	Liter per minute
UHC	=	Unburned hydrocarbons
Z	=	Axial distance measured from the swirler flare exit

I. Introduction

Pollutant emissions emanating from gas turbine engines has been of great concern in the last three decades. These emissions, including nitrogen oxides, carbon monoxide and unburned hydrocarbons, has a great impact on the environment. The reduction of CO, NO_x and UHC simultaneously is challenging due to the fact that the basic feature of nearly all methods of emissions reduction is a "trade-off" between CO and UHC in one side and NO_x on the other side.[1] The introduction of swirling flow in the gas turbine engine industry proved a promising way in reducing pollutant emissions. Highly swirling flows create an internal flow recirculation zone that entrains and circulates a portion of the hot combustion products in the gas turbine combustor dome. This recirculation zone does not act only as a heat reservoir but it plays an active role in better mixing, improving blowout limits, reducing the combustor length and lowering emissions.[2-4] Continuous efforts to further decrease NO_x levels require an understanding of the physical and chemical processes that occur within the recirculation zone. It was found that increasing the swirl intensity (provided that the recirculation zone exists) does not significantly affect NO_x levels.[5,6] The effect of the recirculation zone on the NO_x formation was previously observed, but different conclusions were

* Graduate Student, Department of Aerospace Engineering and Engineering Mechanics, ML 0070, University of Cincinnati, Cincinnati, OH 45219, AIAA Student Member
† Professor, Department of Aerospace Engineering and Engineering Mechanics, ML 0070, University of Cincinnati, Cincinnati, OH 45219, AIAA Member.
‡ Manager, Advanced Combustion Engineering, GE Aircraft Engines, One Neumann Way, MD E404, Cincinnati, OH 45215, AIAA Fellow.

obtained. For example, Mellor et al,[7] measured the temperature and species within a combustor using liquid propane at an overall equivalence ratio of 0.217. They observed that high NO concentrations were found along the centerline in the region thought to be within the recirculation zone. Chen ,[6] concluded that for equivalence ratios ≤ 0.5, NO_x is formed in the recirculation zone. Gupta et al,[8] observed that the combustion flame length was shorter than the length of their respective toroidal recirculation zones. Measured NO concentrations were the highest in the toroidal recirculation zones. Unfortunately, the equivalence ratios used in their study was not mentioned. Claypole et al,[9] studied the influence of swirl levels on the aerodynamics and NO_x emissions at a constant load. The fuel used was natural gas with 10 % excess air above stoichiometry. It was concluded that the recirculation zone, despite its long residence time and moderate temperatures, did not contribute to NO_x formation. Kenbar et al,[5] studied the effect of swirl on the NO_x formation within a gas fired furnace. It was concluded that most of the NO_x formation took place in the reaction zone. Some further reactions occur within the recirculation zone.

From these studies, one can predict that different modes of NO_x formation happen within the recirculation zone based on the overall equivalence ratio.

A great number of studies considered only exit plane measurements and the combustion processes were considered as an in/out process. On the other hand, only few studies within the combustor have been reported. To understand pollutant emissions formation/destruction processes, the need for in-situ species measurements becomes increasingly important. LDV and PDPA measurements have been utilized in the Combustion Diagnostics Lab (CDL) at the University of Cincinnati to study the flow aerodynamics of swirl stabilized flames and the recirculation zone interactions. In this study, species concentration measurements within the combustion chamber are employed in efforts to combine the chemical processes with the flow aerodynamics. Measurements of species concentrations including CO_2, CO, H_2O, O_2, NO, NO_2, SO_2 and CH_4 are taken within the recirculation zone and the chamber centerline at different equivalence ratios. To study the effect of the inlet air temperature at a wide range of equivalence ratios, exit plane measurements are also addressed. To determine the locations where the formation of pollutant emissions is a maximum, no dilution air was admitted to the chamber in order to isolate its effect on the pollutants formation/destruction processes.

II. Experimental setup

The experimental configuration used in the present study consists of a co-annular counter-rotating swirl cup assembly, a simplex nozzle and a square chamber. A multiple gas analyzer based on Fourier transform infrared (FTIR) spectroscopy is employed to measure the emission species concentrations generated during combustion of Jet-A fuel. In addition, a paramagnetic O_2 analyzer is used to measure the excess oxygen within the sample. Platinum fine wire Pt/Pt30Rh type B thermocouple is utilized to obtain a spatially resolved temperature profile.

A. Test Facility

A schematic diagram of the test facility is shown in Fig. 1. The air system consists of a compressor, a dryer, a 0.5 micron particle filter, a pressure regulator, and an air storage tank. An ASME orifice flow meter is utilized to measure airflow rate into the system. A 39 kW heater can be used to preheat the air per test requirements. The air passes through a honeycomb flow conditioner to the test section. The fuel system includes a fuel tank, a fuel pump, a fuel filter, a coriolis flowmeter, and a needle valve for controlling the fuel flow rate.

The fuel is injected by a hollow cone simplex atomizer with 90-deg spray angle mounted in the center of the swirler. The swirler used in this study is designed to generate two coaxial, counter swirling airstreams. The swirler flare exit diameter D is 40mm. The detailed design of this swirl cup has been fully described elsewhere.[10] The combustion chamber used in this study is made of both quartz glass and stainless steel plates with a 3.0 inch square cross-section and a height of 14 inches. The chamber is modified such that it can allow a gas sampling probe to traverse in two dimensions inside it.

B. Gas sampling probe.

Gas samples were extracted from the combustion chamber using a gas sampling probe with active liquid cooling jacket. Kent et al,[11] referred the greatest sampling errors arise from probe size and shape effects to radial shifts in stream lines. Gouldin,[12] suggested that if the probe and tip diameters are smaller than 10% of any characteristic length scale of the flow provided that the probe is aligned with the mean flow for at least 10 diameters, interference effects can be kept minimum. As a result, the sampling probe is designed with as small diameter as possible such that flow perturbations will be minimized. Moreover, the probe configuration is based on standard designs in order

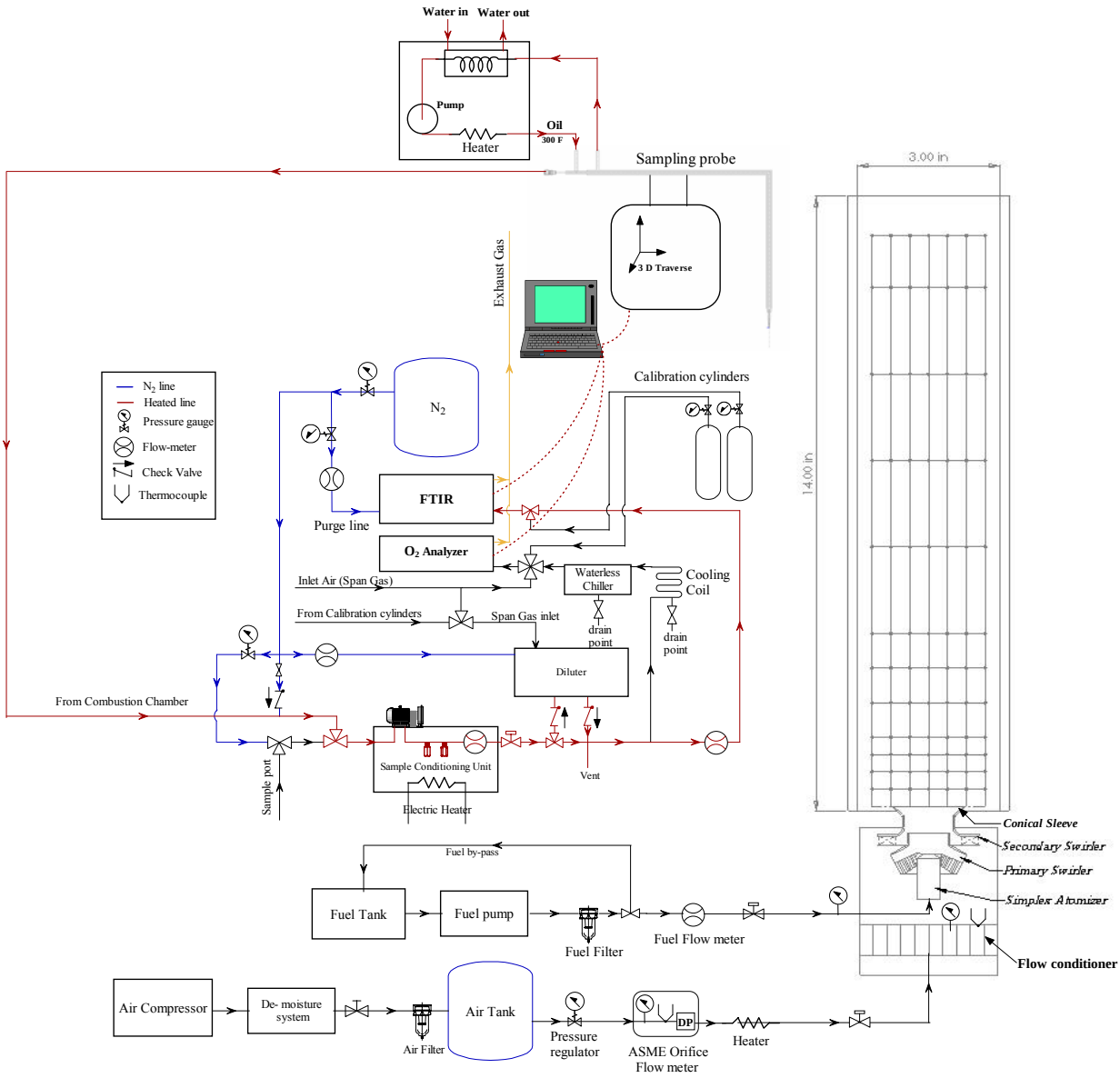


Figure 1. Schematic diagram of the test facility.

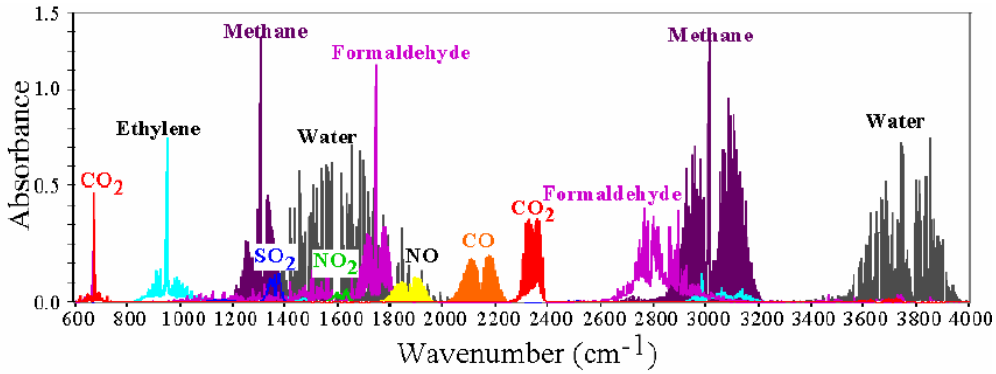


Figure 2. Absorbance spectrum of typical species found in a gas turbine combustor.[17]

to align the tip with the mean flow. The sampling probe is made of stainless steel with an outside dimension of 0.25 inch and a tip inlet diameter of 0.027 in. The probe consists of three concentric tubes. The inner most tube is used for the gas sample suction while the other two tubes are used for the circulation of the cooling fluid. Heated Teflon-lined stainless steel tubing is utilized to deliver the sample to the gas analyzers. The temperature in the tubing was kept at 150°C to avoid water vapor condensation. In order to freeze the reactions in the sampling probe, water cooling was circulated in the probe to cool the sample and protect the probe itself from excessive heating. Uncertainties in the measured water vapor and nitrogen oxides concentrations in the sample were significant. Similar observations were reported.[7,13] This might be attributed to the water vapor condensation on the sampling tube surface due to excessive cooling.[14] In order to eliminate this problem, oil is utilized instead of water to cool the sample. It was suggested that cooling the sample down to a temperature of 300 °C is sufficient to freeze further reactions in the probe.[15] The oil cooling system was designed such that the oil inlet temperature is to be kept at 225 °F. This will be sufficient enough to cool the sample to temperatures below 300 °C and keep the water in the sample in its vapor phase. Hassan et al,[16] suggested that the influence of the velocity appears to be small in the range of 40-100 m/s. As a result, the flow rate of the gas sample is kept at moderate values as compared with the flow velocities. Repeatability of the measurements at different days for CO and NO was in agreement within 3.0%. This agreement made us confident about the current configuration.

C. Emission measurements

At each measurement location, primary species of interest including carbon dioxide, carbon monoxide, nitrogen dioxide, nitrogen monoxide, sulfur dioxide, water vapor, and unburned hydrocarbons are measured. These measurements are employed using a MultiGas 2030 FTIR analyzer. Particles in the sample are removed by passing the sample through a Sample conditioning unit that includes two heated particle filters to remove particles larger than 0.6 μm as shown in Fig 1.

The choice of a multiple gas analyzer based on Fourier Transform Infrared (FTIR) to be used for gas concentration measurements over single gas analyzers was preferred due to many reasons, among these are the ability of the FTIR analyzer to measure on a wet basis makes it possible to measure the species concentrations with minimal sample conditioning. Besides its ability to measure the species of interest like CO, CO_2, NO, NO_2, and H_2O, the FTIR can also measure many other species like sulfur dioxide, methanol, ethylene and formaldehyde. Another benefit of the FTIR is its ability to perform data validation, since the infrared spectrum recorded is stored along with the concentration report. This means that any unusual data points can be reevaluated or validated long after the test was completed.[17]

FTIR is based on the fact that every molecule (except atomics such as noble gases and homonuclear diatoms which are infrared inactive) absorbs infrared radiation with a characteristic, molecule-specific absorption spectrum. Gas species can be identified and quantified based on the location and magnitude of these absorptions. Figure 2 represents the absorbance spectrum of different gases that can be found in a gas turbine combustor. The basis of all quantitative analysis in FTIR is Beer's Law, which relates concentration to absorbance.[18]

Since FTIR analyzers do not measure diatomic gases (e.g. O_2 and N_2), a paramagnetic O_2 analyzer is employed to measure the excess Oxygen within the sample.

D. Temperature measurement

Mean temperatures are obtained using Pt/Pt30Rh type B bare wire thermocouple with a wire diameter of 0.03 in. Catalytic action is reduced by coating with a layer of silica using a flame incorporating silicone oil.[19] The thermocouple bead diameter is estimated to be about 0.007 inch.

E. Test Conditions

The goal of this study is to understand the chemical processes along with flow aerodynamics in the primary zone of a model gas turbine combustor without dilution air present. Two main parameters are varied; these are the overall equivalence ratio and the inlet air temperature. Throughout the measurements, the pressure drop across the combustor is kept constant at 4.0 ±0.1% of the atmospheric pressure. Equivalence ratio variations are done by varying the fuel flow rate. Thus, the effect of the residence time can be isolated from the equivalence ratio variations. Exit plane measurements were utilized to study the effect of varying both the inlet air temperature and the overall equivalence ratio on pollutant emissions of interest such as carbon monoxide and nitrogen oxides. These measurements were executed at the center point, 13.0 in above the swirler flare exit. Inlet air was preheated to three different temperatures; these are 130 °F, 300 °F, and 450 °F. The equivalence ratio was varied from near lean blowout condition to a rich fuel state at $\phi = 1.2$. Two sets of species concentration measurements within a model combustor have been taken. In the first set, samples were taken at a total number of 84 locations to represent both

longitudinal and radial profiles. Three equivalence ratios have been tested at ambient inlet air temperature. These equivalence ratios are 0.45, 0.7, and 0.95. The last set included measurements at the centerline with more detailed measurements in the recirculation zone.

III. Results and Discussion

A series of measurements at the chamber exit plane have been executed to study the effect of inlet air temperature at different equivalence ratios. The measurements include carbon monoxide, nitric oxide, nitrogen dioxide, methane, and other major species. Temperature was measured using Pt/Pt30Rh type B thermocouple. Chemical composition has been utilized to determine combustion gas temperatures. Williamson et al,[20] discussed the determination of jet engine combustor temperature using thermocouples and by using gas analysis. For temperatures above 2800 °F, gas analysis has been more satisfactory than thermocouple measurements. Sawyer,[21] also employed this approach for temperature determination and concluded that the results confirm the advantages of this approach over thermocouple measurements which suffer difficulties of probe integrity in the combustor environment and radiation correction.

A. Exit plane measurements

The temperature profiles at different equivalence ratios for different inlet air temperatures are shown in Fig.3. A comparison between the thermocouple measurements (without corrections) and temperatures calculated from gas analysis confirms the benefit of this approach. The difference between the temperature measured and the temperature calculated is about 150-200 K. This difference can be attributed to thermocouple measurements without correction for conduction, radiation and other losses as was discussed elsewhere.[22-25] In lean and near stoichiometric flames, increasing the inlet air temperature increases the combustion temperature. The maximum combustion temperatures are attained on the fuel lean side of stoichiometry with increasing the inlet air temperature. On the contrary, increasing the inlet air temperature in the rich fuel side leads to a decrease in combustion temperature.

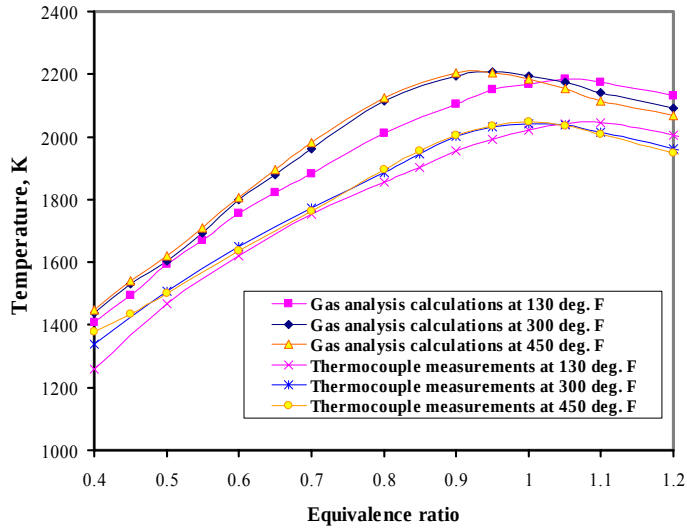


Figure 3. Combustion Temperature profiles at different inlet air temperatures for different equivalence ratios

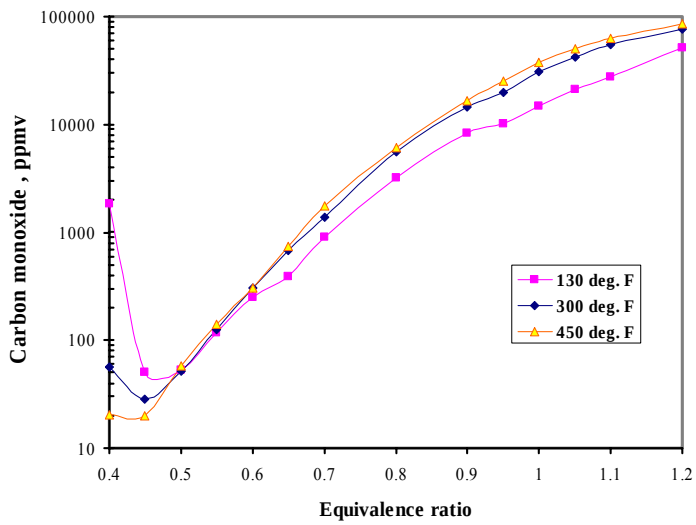


Figure 4. Effect of inlet air temperature on Carbon monoxide concentrations, ppm, at different equivalence ratios.

Figure 4, represents the effect of inlet air temperature on the concentrations of CO at the center point, 13.0 inch above the swirler exit, at different equivalence ratios. The CO formation/destruction process can be divided into three regions based on the equivalence ratio. The first region where the equivalence ratio is up to $\phi = 0.45$. In this region it is clear that by increasing the inlet air temperature or the equivalence ratio, CO concentrations will decrease significantly by enhancing the oxidation to CO_2. The minimum measured CO found in the exhaust was at an equivalence ratio $\phi = 0.45$ for all inlet air temperatures. At low equivalence ratios ($\phi < 0.45$) and low inlet air temperatures, the combustion temperature is not high enough to accelerate the oxidation of CO to CO_2. This can be

attributed to the so called thermal quenching process as reported by Sawyer et al.[13] A temperature is reached in the chamber at which further oxidation reaction of CO to CO_2 effectively stops, and exit CO emissions are fixed. Bowman,[26] related the quenching of CO at lower temperature to the depletion of radical species, such as OH, by recombination reactions. From Fig.3, the thermal quenching temperature for CO was at about 1500 K.

In the second region, ($0.5 < \phi < 0.6$), increasing the equivalence ratio increases the amount of CO measured at the exit. However, the effect of increasing inlet air temperature is less pronounced. In the second region, the combustion temperature is higher than the thermal quenching temperature of CO but still less than the temperature where dissociation process starts significantly.

In the third region, ($\phi > 0.6$), CO production by chemical dissociation of CO_2 becomes significant. Generally, increasing the inlet air temperature in this region leads to an increase of CO. At rich fuel air ratios, higher amounts of CO are produced to the extent that increasing inlet air temperature at these rich equivalence ratios will lead to a decrease in the combustion temperature, as shown in Fig.3.

In this study, the measured nitric oxide and nitrogen dioxide are coupled together and expressed in terms of oxides of nitrogen (NO_x). The effect of inlet air temperature on oxides of nitrogen measured near the chamber exit at different equivalence ratios is shown in Fig.5. In the range of equivalence ratios of $0.5 < \phi < 1.0$, a significant increase in the amount of NO_x is produced with increasing inlet air temperature. However, in the rich fuel side, the NO_x emissions were found to decrease with increasing inlet air temperature. This may be attributed to the strong coupling between the thermal nitrogen oxide formed and the combustion temperature. Maximum NO_x concentrations were achieved on the fuel lean side of stoichiometry just upstream of the location of maximum combustion temperature.

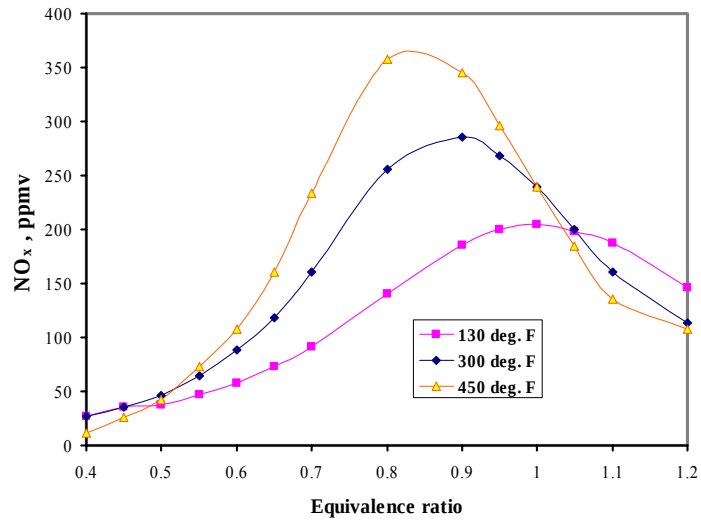


Figure 5. Nitric oxides (NO_x) concentration at the exit plane at different equivalence ratios

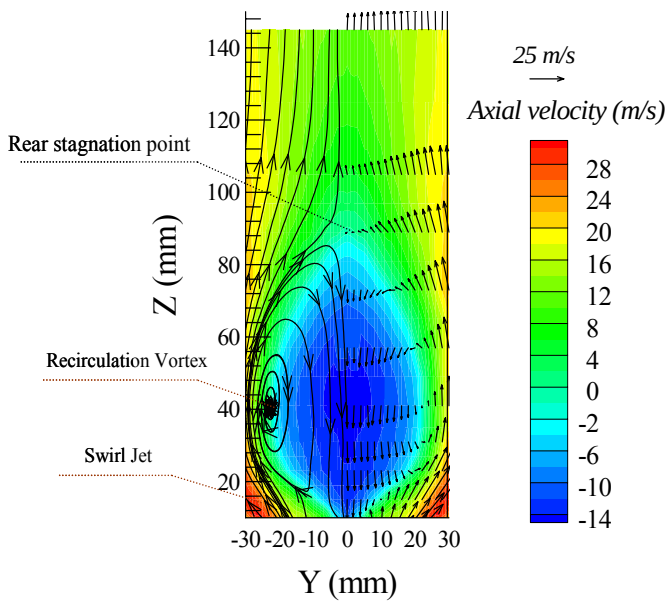


Figure 6. Velocity vectors, Axial velocity contours and stream lines at an equivalence ratio $\phi = 0.7$. [27]

B. Species measurements within the combustor

Measurements in 12 lines within the model chamber are performed to form a plan. These lines are at 10, 20, 30, 40, 60, 80, 100, 150, 200, 250, 300 and 330mm above the swirler flare exit, as shown in Fig.1. Within every line, 7 points are measured. Using ambient inlet air, three equivalence ratios have been chosen for comparison. The first equivalence ratio is at $\phi = 0.45$ which produces the minimum CO and low NO_x at the exit plane. The second one is at an equivalence ratio $\phi = 0.7$, where velocity measurements and spray characteristics have been studied previously.[27] The last one is at an equivalence ratio of $\phi = 0.95$ which represent the case of maximum NO_x produced in the lean side of stoichiometry.

To combine the understanding of flow aerodynamics and thermal processes within a chamber, LDV measurements of gas velocities within the model combustor used in this study are presented in Fig. 6. The swirling flow emanating from the swirler remains attached to the 45° conical sleeve until it impinges on the chamber walls. Due to the high swirl velocity a central toroidal recirculation zone forms. Flame visualization showed that fuel air mixture and products of combustion are forced to flow around this recirculation zone and then entrained near its end. Similar observations were reported by Chen et al.[28] The height of the recirculation zone can be determined by the rear stagnation point which is the location of the zero axial velocity at the centerline. In this study, the flow field inside the combustion chamber will be divided to three main regions, these are the swirling jet region where reacting fuel air mixture flow around the recirculation zone, the second region is the recirculation zone, and the third region is the post recirculation zone.

Fu et al,[27] studied the effect of the equivalence ratio variations on spray combustion using the same combustor model configuration employed in this study, it was observed that the size of the recirculation zone does not change as much by changing the equivalence ratio. Thus, we can use the information about the recirculation zone height that was determined for the case of $\phi = 0.7$ and apply it for the other equivalence ratio cases. The recirculation zone was determined to end at about $Z/D \approx 2.25$.

Contour maps of nitric oxides, carbon monoxide, local equivalence ratio, and the flame temperature for $\phi = 0.45$ are presented in Fig. 7. The local equivalence ratio and the flame temperatures are calculated using the gas analysis data. The majority of NO_x formed is found to be inside the recirculation zone, Fig.7.a. Within the recirculation zone itself, the maximum concentration of NO_x is found in and around the recirculation vortices where the local equivalence ratios are near stoichiometric and the temperature is high enough to promote NO_x formation. However, at the rear stagnation point of the recirculation zone, NO_x is found with lower concentrations. This can be attributed to the entrainment of the swirling air into the recirculation zone especially near the rear stagnation point. Higher amounts of measured O_2 is observed in this area. This

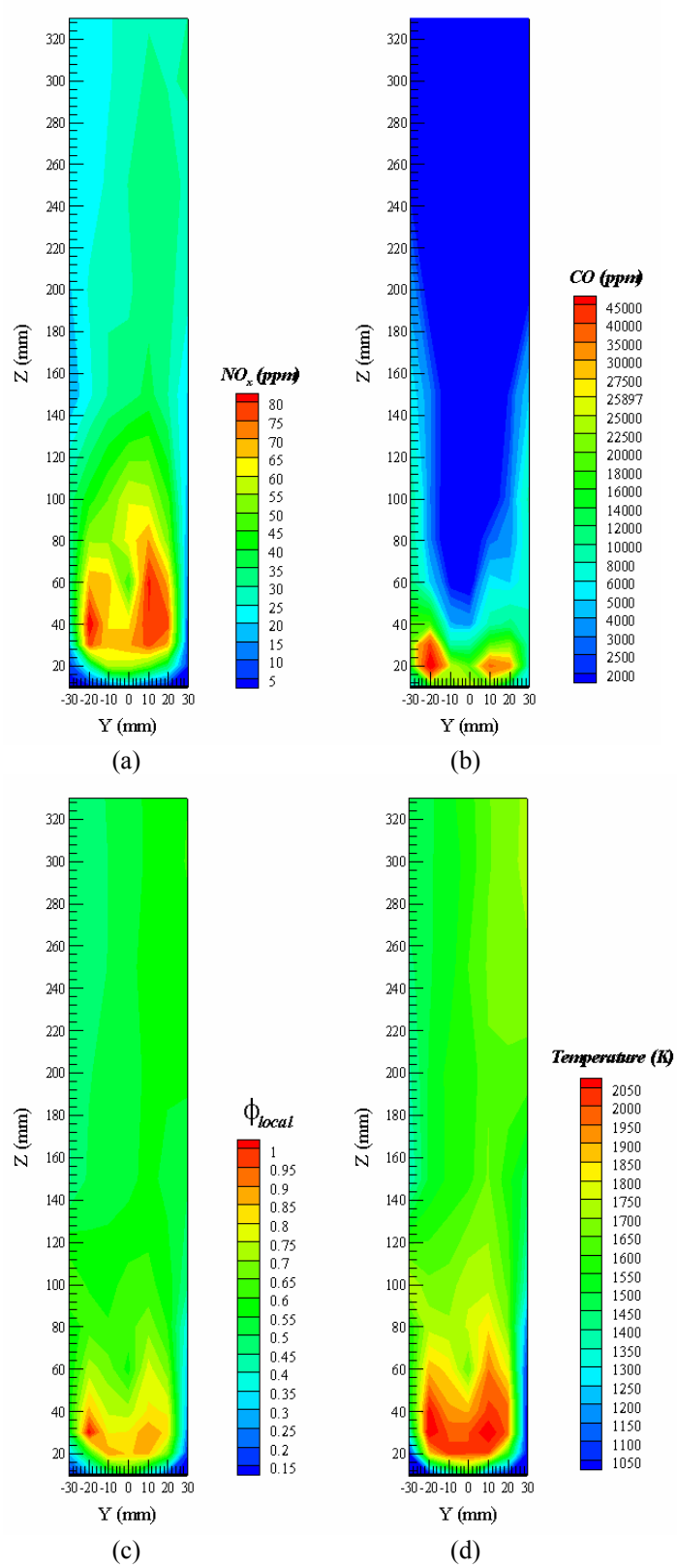


Figure 7. Spatial distribution of measured concentrations at an equivalence ratio ϕ = 0.45: (a) NO_x, (b) CO, (c) Local equivalence ratio, (d) Temperature (K)

entrained air helps in decreasing the local equivalence ratio from near stoichiometry to around 0.6, leading to a lower combustion temperature and thus lower NO_x production near the recirculation tip. At this equivalence ratio ($\phi = 0.45$), the flame is short compared to other cases of higher equivalence ratios. NO_x levels reach a stabilized state with little changes in the measured concentrations above Z/D ≈ 4 where D is the swirler flare exit diameter.

Carbon monoxide concentration increases rapidly in the flame zone to a maximum value. At Z/D = 0.25, which is just downstream of the forward stagnation point, the maximum CO concentration measured is found in the centerline. This indicates that a portion of the fuel emanating from the fuel nozzle impinges on the recirculation zone and produces high amounts of CO. At Z/D = 0.5, the maximum CO concentration measured is at the recirculation vortices, the amounts of CO produced at this section is the highest. At the center of the recirculation zone, CO concentration decreases due to the oxidation to CO_2. At Z/D=0.75, air dilution and oxidation chemical reaction to CO_2 become more significant especially inside the recirculation zone. This is confirmed by the measured CO_2 at this section where it attained its maximum concentrations. At Z/D > 1.0, CO Concentration becomes higher near the chamber walls. This is due to the relatively low temperature near the wall that decelerates the conversion of CO to CO_2 , as shown in Fig.7.d, It is now clear that at $\phi = 0.45$, the combustion is completed within the recirculation zone. This result is in agreement with Chen et al.[28] who concluded that for overall equivalence ratios below 0.5, the combustion is completed in the recirculation flow region. The recirculation zone plays an active role in the mixing process by the simultaneous entrainment of fuel air mixtures and delivery of combustion products instead of being just an aerodynamic blockage filled with hot combustion products. Also, it can be concluded that for this equivalence ratio, the recirculation is the main producer for NO_x due to its higher temperature and local equivalence ratios as compared with the post recirculation zone.

The second case studied is at $\phi = 0.7$ where longer flame is observed, the recirculation zone volume is not enough to burn all the fuel inside it, and the flame is extended downstream of the recirculation zone. As shown in Fig. 8.a., the maximum NO_x concentration produced is in the

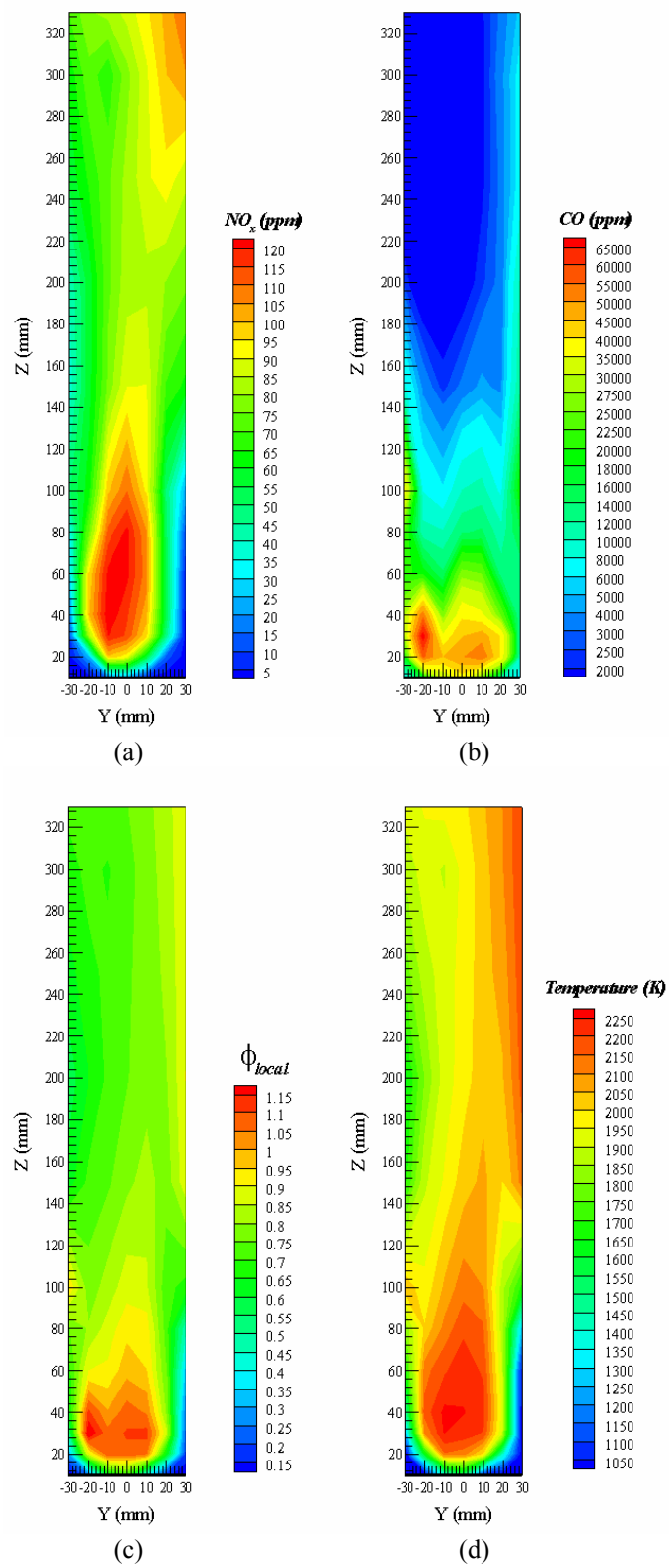


Figure 8. Spatial distribution of measured concentrations at an equivalence ratio ϕ = 0.7: (a) NO_x, (b) CO, (c) Local equivalence ratio, (d) Temperature (K)

recirculation zone along the centerline not in the circulation vortices like the case of $\phi = 0.45$. Since the flame is extended in front of the recirculation zone, the entrained mixture at the rear stagnation point will be at very high temperature and chemically reactive. It will continue to react in the recirculation zone, resulting in an increase in the local equivalence ratio to near stoichiometric and to high combustion temperatures. Finally, this will result in an increase of the NO_x production within the recirculation zone.

Generally, CO produced in this case is higher than that of $\phi = 0.45$. Highest concentrations are observed inside the recirculation zone at Z/D = 0.5. This is attributed to the increased fuel penetration to the recirculation zone with increasing the equivalence ratio. Unlike the case of $\phi = 0.45$, higher concentrations of CO are found at the centerline not around the recirculation vortex. This confirms the observation of the entrained reactants at the recirculation tip and higher amounts of CO produced since the temperature is higher than 1800 K and dissociation process is likely to be more pronounced. CO concentration decrease, after attaining its maximum, slowly farther downstream and reach a quasi-steady state at about Z/D ≈ 6.

Figure 9 represents the last case studied where the overall equivalence ratio is set to $\phi = 0.95$. Relatively cold fuel air mixture flows around the recirculation zone. The local equivalence ratio of this mixture increases due to the continuous reaction until it reaches stoichiometric just above the recirculation rear stagnation point. This stoichiometric mixture is entrained into the recirculation zone where oxygen is depleted. The local equivalence ratio inside the recirculation zone become in the rich fuel side. Relatively lower amounts of NO_x concentrations are produced as compared with the post recirculation zone. However, these concentrations in the recirculation zone are higher than that of the case of $\phi = 0.45$. Downstream of the recirculation zone, NO_x continues to form and attains its maximum value where the local equivalence ratio is near stoichiometric. The quasi-steady state for NO_x does not occur within the chamber since the flame is longer than the chamber.

The maximum concentration of CO is inhibited in the recirculation zone. The trend is similar to the case of $\phi = 0.7$, but the concentrations are higher by one order of magnitude at the exit plane.

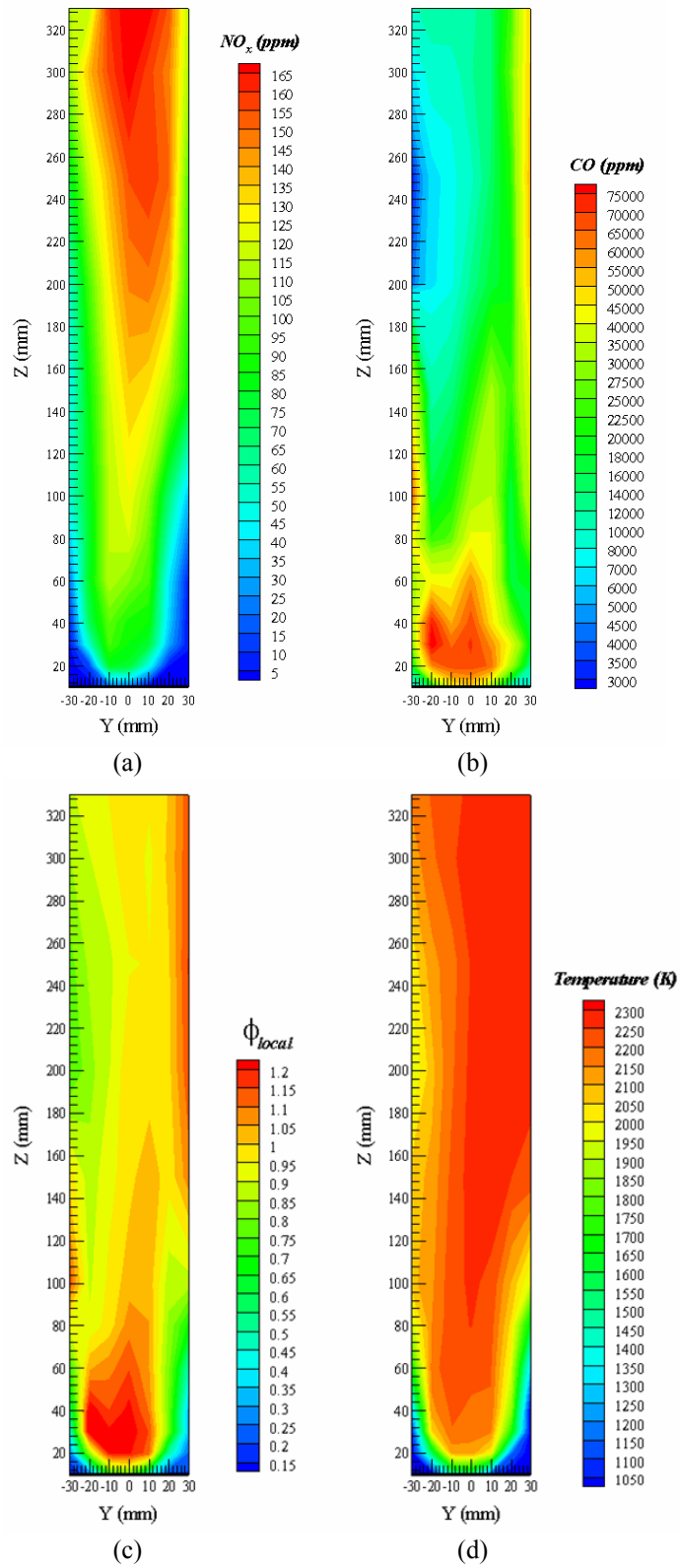


Figure 9. Spatial distribution of measured concentrations at an equivalence ratio $\phi = 0.95$: (a) NO_x, (b) CO, (c) Local equivalence ratio, (d) Temperature (K)

C. Centerline species measurements

More measurements have been conducted in the chamber centerline at the equivalence ratios of

interest. A comparison between these three cases for the local equivalence ratio, combustion temperature, CO, UHC and NO_x is presented in Fig. 10. Increasing the overall equivalence ratio will generally lead to an increase in the local equivalence ratio and combustion temperature. For the case of $\phi = 0.45$, the maximum local equivalence ratio reaches about 0.85. At this point, the amount of CO produced is much higher than the equilibrium CO. similar trends are observed for the other equivalence ratios. CO concentration increases rapidly in the flame zone to a maximum value. Following this maximum, CO concentration decreases in a relatively slower rate compared to its formation to the final exit plane value. These results confirm the suggestion that both the formation and destruction of CO is kinetically controlled. Bowman[26], reported that for hydrocarbon flames, the principle CO oxidation reaction is

$$CO + OH \rightleftarrows CO_2 + H \quad (1)$$

The rate coefficient of reaction 1 exhibits significant temperature dependence at combustion temperatures in excess of the thermal quenching temperature (1500 K).

The temperature profiles for both overall equivalence ratios of $\phi = 0.45$ and $\phi = 0.7$ exhibit the same trends. At $\phi = 0.45$, the maximum temperature obtained is at Z/D ≈ 0.625, where the maximum local equivalence ratio is also obtained. For the case of $\phi = 0.7$, the location of maximum temperature is farther downstream at Z/D ≈ 1.0 where the local equivalence ratio is about 1.1. For both cases, the temperature decreases after the maximum is obtained. For the case of $\phi = 0.95$, the trend is different. The maximum equivalence ratio obtained in the centerline is about 1.2. The centerline temperature in the recirculation zone is lower than that of $\phi = 0.7$ due to oxygen depletion and the incomplete combustion at this rich fuel state. The temperature continues to increase as the local equivalence ratio falls towards the stoichiometric condition.

The UHC in this study is represented by the methane (CH_4) that is found in the gas analysis. The trends of CH_4 formation/destruction are very similar to that of CO. It is noteworthy that the locations of maximum CH_4 are found to precede that of the carbon monoxide. The maximum NO_x concentrations are found to be farther downstream of CO and UHC maximums. Similar observations were reported by Sawyer,[21] who concluded that since these emissions exhibit temperature dependence, the possibility exists for tailoring the combustor temperature-time characteristics to reduce all three species simultaneously.

Profiles of nitrogen oxide emissions at the centerline are presented in Fig. 10.e. At equivalence ratios of $\phi = 0.45$ and $\phi = 0.7$, the maximum NO_x concentrations are found inside the recirculation zone. Following this maximum, NO_x concentration continues to decrease to a fixed concentration inside the chamber. In the case of $\phi = 0.95$, NO_x concentration increases all the way till the exit plane, and a quasi-steady state is never obtained.

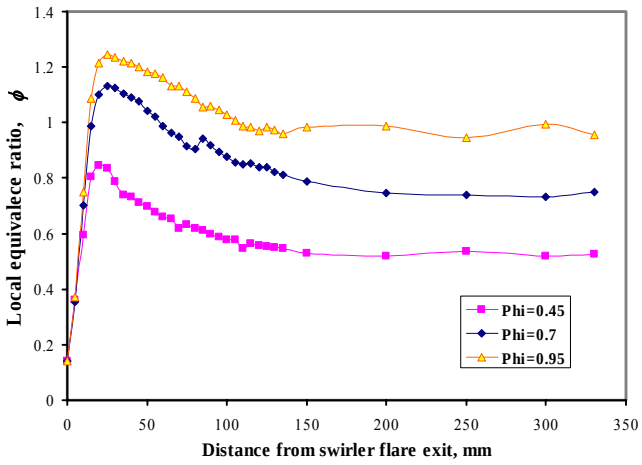
IV. Conclusion

At very lean equivalence ratios, if the combustion temperature is less than the quenching temperature (1500 K), increasing either the inlet air temperature or the equivalence ratio will accelerate the oxidation of CO to CO_2 resulting in a reduction of CO emissions at the exit plane. But, if the combustion temperature is higher than the quenching temperature, then increasing the inlet air temperature will lead to an increase in the production of CO by dissociation.

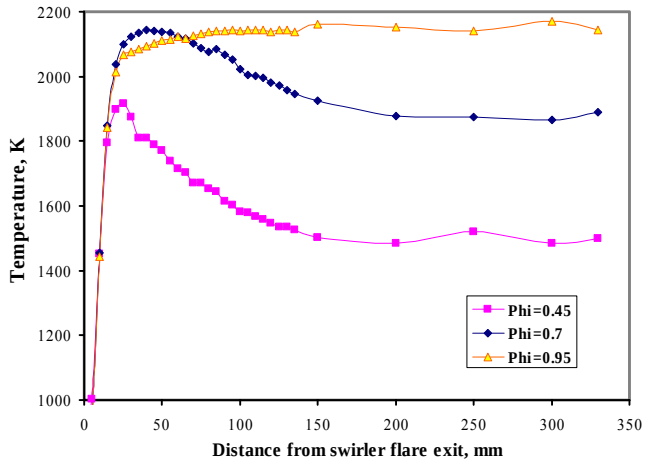
The locations of maximum UHC are found to precede that of the CO. The maximum NO_x concentrations are found to be farther downstream of CO and UHC maximums. Thus the possibility exists that these emissions can be lowered simultaneously by controlling the temperature within the chamber.

NO_x emissions increase with increasing the inlet air temperature and attain its maximum on the lean side of stoichiometry. In the rich fuel combustion, increasing inlet air temperature will help reducing NO_x production

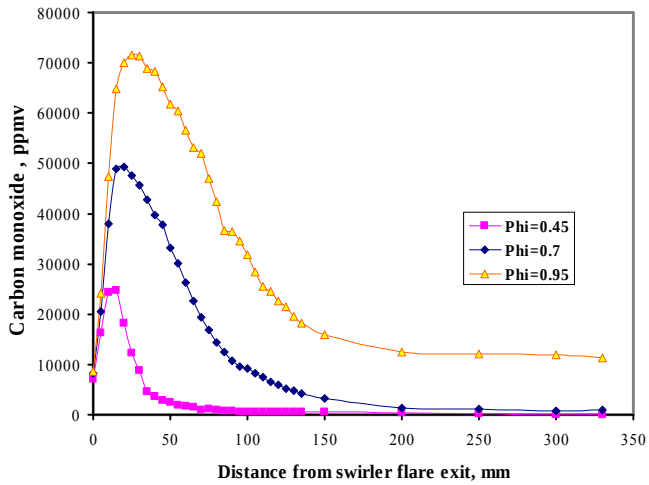
In swirl stabilized flames, NO_x, production is believed to be a function of both the recirculation zone and the flame length. With equivalence ratio variations, it is observed that NO_x production can be divided into three modes. In the first one, which is at low overall equivalence ratios up to $\phi = 0.6$, the flame length is shorter than the recirculation zone height. Most of the combustion is believed to be completed within the recirculation zone. Thus, the recirculation zone becomes the main producer of NO_x due to its higher temperature and local equivalence ratios as compared with the post recirculation zone. The second mode, which is a transitional one, appears at overall equivalence ratios $0.6 < \phi < 0.9$. In this mode, the local equivalence ratio in the recirculation zone is near stoichiometric. However, the flame is longer than the recirculation zone height. NO_x production in this mode is shared between the recirculation and the post recirculation zones. In the last mode, at equivalence ratios $\phi > 0.9$, the recirculation zone is fuel rich. The post recirculation zone appears to be the main producer of NO_x.



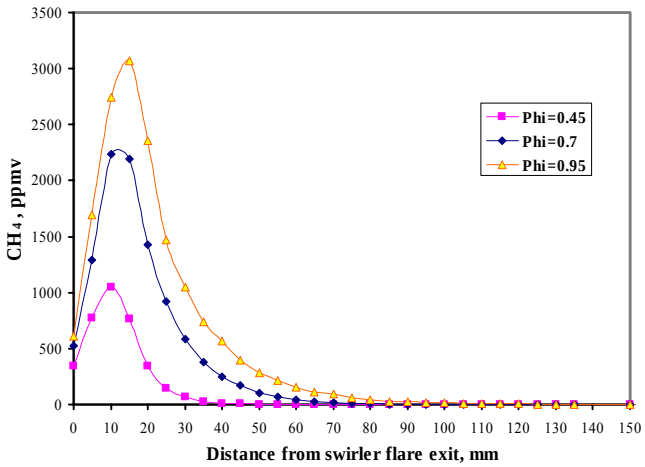
(a)



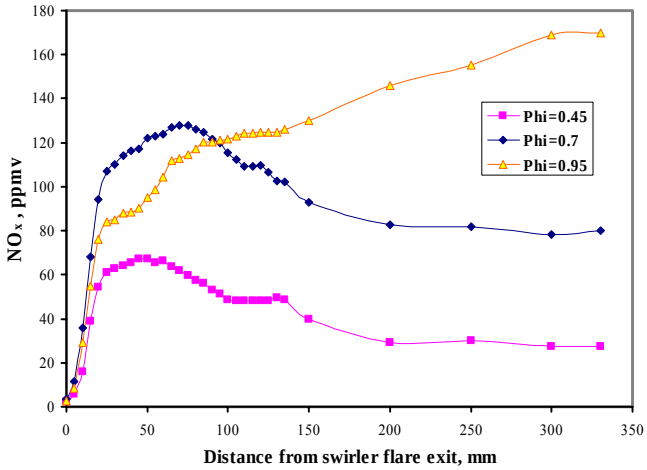
(c)



(b)



(e)



(c)

Figure 10. Centerline profiles of (a) Local equivalence ratio, (b) Temperature, (c) Carbon monoxide (d) CH_4, and (e) NO_x using Jet-A fuel at different equivalence ratios, ambient inlet air

References

[1]Lefebvre, A.H., "Pollutant Control in Continuous Combustion Engines," *Proc. 15th Symposium (intl.) on Combustion*, The Combustion Institute, Pittsburgh, PA, 1975, pp. 1169–1180.

[2]Lefebvre, A.H., *Gas Turbine Combustion*, 2nd ed., Taylor and Francis, 1998

[3]Beer, J.M. and Chigier, N.A., *Combustion Aerodynamics*, Applied Science Publishers LTD, London, 1972

[4]Syred, N., Beer, J.M., "Combustion in Swirling Flow: A Review," *Combustion and Flame*, Vol. 23, 1974, pp. 143-201.

[5]Kenbar, A.M.A., Beltagui, S.A., Ralston, T., and Maccallum, N.R.L., "Measurement and Modeling of NOx Formation in a gas fired furnace", *Combustion science and Technology*, 93, 1993, pp.173-192

[6]Chen, R.H., "Some Characteristics of NO_x Emission of Hydrogen-Air Flames Stabilized by Swirl-Generated Flow Recirculation," *Combustion Science and Technology* 110-111, 1995, pp. 143-161

[7]Mellor, A.M., Anderson, R.D., Altenkirch, R.A., and Tuttle, J.H., "Emissions From and Within an Allison J-33 Combustor," *Combustion Science and Technology* 6, 1972, 169-176.

[8]Gupta, A. K., Syred, N., and Beer, J. M., "Fluctuating Temperature and Pressure Effects on the Noise Output of Swirl Burners," *Proc. 15th Symposium (intl.) on Combustion,* The Combustion Institute, Pittsburgh, PA, 1975, pp. 1367–1377.

[9]Claypole, T.C., and Syred, N., "The Effect of Swirl Burner Aerodynamics on NO_x Formation, " *Eighteenth symposium (International) on combustion*, The Combustion Institute, Pittsburgh, PA, 1981, pp. 81-89.

[10]Mongia, H.C., Al-Roub, M., Danis, A., Elliot-Lewis, D., Johnson, A., Vise, S., Jeng, S-M., McDonell, V.., and Samuelsen, G.S., "Swirl Cup Modeling Part 1, " *AIAA Paper,* 2001-3575, 2001.

[11]Kent, J.H., and Bilger, R.W., "Measurement Techniques in Turbulent Diffusion Flames, " *First Australian Conference on Heat and Mass Transfer*, Monash University, Melborne, Sec. 4.4, 1973, pp. 39-46.

[12]Gouldin, F.C., "Testing and Measurement Techniques in Heat Transfer and Combustion, " *AGRAD CP-281*, 1980

[13]Sawyer, R.F., Teixeira, D.P., and Starrkman, E.S., "Air Pollution Characteristics of Gas Turbine Engines, " *ASME Trans., J. Eng. Power* 91, 1969,pp.290-296.

[14]Allen, J. D. "Probe Sampling of Oxides of Nitrogen from Flames, " *Combustion and Flame*, 24, 1975,pp. 133-136

[15]Chedaille, J. and Braude, Y. *Industrial Flames. Vol. 1, Measurements in Flames*. Edward Arnold Publishers, London, 1972.

[16]Hassan, M.M.A., Lockwood, F.C., Moneib, H.A., "Fluctuating Temperature and Mean Concentration Measurements in a Vertical Turbulent Free Jet Diffusion Flame, " *Rivista dei Combustibili*. v 34, n 9-12. Sep-Dec 1980, pp. 357-372

[17]Marran, D., Kenny, P., Markham, J., Jalbert, P., Moyers, R., and Gardner, D., "The Application of FT-IR Spectroscopy to Turbine Engine Exhaust Monitoring, " *21st AIAA Ground Test Conference*, Denver, CO (June 19-22, 2000). Paper AIAA 2000-2211.

[18]Griffithd, P.R. and De Haseth, J.J., *Fourier Transform Infrared Spectroscopy*, New York , NY, Wiley InterScience, 1986

[19]Frisrom, R. M., *Flame Structure and Processes*, Oxford University Press, Oxford, UK, 1995.

[20]Williamson, R. C., and Stanforth, C. M., "Measurements of Jet Engine Combustion Temperature by The Use of Thermocouples and Gas Analysis, " *SAE paper No 690433*, Nat. Air Trans. Meeting, New York, 1969.

[21]R. F. Sawyer, "Experimental Studies of Chemical Processes in a Model Gas Turbine Combustor, " *Emissions from continuous Combustion systems*, Plenurn Press, New York, 1972., pp. 243-254.

[22]Bradley, D., and Matthews, K.J., "Measurement of High Temperature with Fine Wire Thermocouple, " *Journal of Mechanical Engineering Science*, Vol 10, NO.4, 1968, pp. 299-305.

[23]Attya, A. M.; Whitelaw, J. H., "Velocity, Temperature and Species Concentrations in Unconfined Kerosene Spray Flames." *American Society of Mechanical Engineers (Paper).* 1981.

[24]Samuelsen, G.S., LaRue, J.C. and Seiler, E.T. "Instantaneous Two-Component Laser Anemometry and Temperature Measurements in a Complex Reacting Flow. " *Intl. Symp. on Application. of Anemometry to Fluid Mechanics*, 1984

[25]Heitor, V., and Moreira, A.L.N., "Thermocouples and Sampling Probes for Combustion Studies, " *Progress in Energy and Combustion Science.*, 19, 1993, pp. 259-278.

[26]Bowman, C. T., "Chemistry of Gaseous Pollutant Formation and Destruction, " *Fossil Fuel Combustion—A Source Book*, eds., W. Bartok, W. and A. F. Sarofim, Wiley, New York, NY.

[27]Fu, Y., ELKady, A.M., and Jeng, S.M., "Fuel and Equivalence Ratio Effects on Spray combustion of a Counter-Rotating Swirler, " *AIAA Paper* 2005-0354, 2005.

[28]Chen, H.R., and Driscoll, J.F., "The Role of the Recirculation Vortex in Improving Fuel-Air Mixing Within Swirling Flames, " *Twenty-Second Symposium (International) on Combustion*, The Combustion Institute, Pittsburgh, 1988, pp. 531-540.